Sec: SR STAR-A&APEX, ALL IITS | DT: 04/01/2026
Time: 3 Hrs | GTM – 08 (TBT-1) | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	C	B	B	C	C	A	C	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	C	D	B	B	A
21	22	23	24	25					
124	13	3	123	123					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	D	B	C	A	A	D	C	D
36	37	38	39	40	41	42	43	44	45
B	C	A	D	C	A	B	B	B	D
46	47	48	49	50					
5	5	0	9	15					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	D	B	A	C	A	C	D	C	A
61	62	63	64	65	66	67	68	69	70
A	C	C	D	D	B	C	B	A	D
71	72	73	74	75					
3	6	81	45	4					

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HINTS
PHYSICS

1. In the battery connected state

$V = V_0$; $E = E_0$ [$V = Ed$]

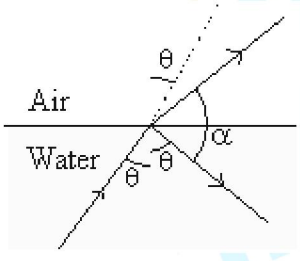
$C = KC_0 \Rightarrow C > C_0$

$Q = CV \Rightarrow Q > Q_0$

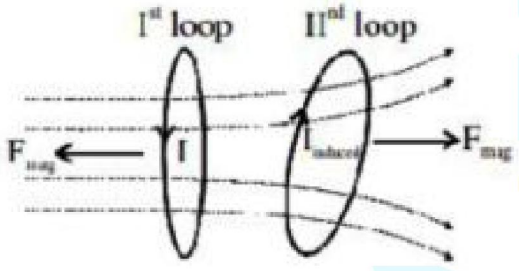
$U = \frac{1}{2}CV^2 \Rightarrow U > U_0$

2. $\theta < C$

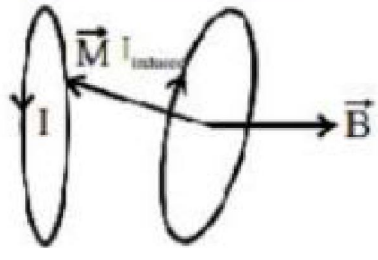
$\alpha < 180° - 2\theta$



3.



When suddenly current injected into the left loop flux through the IInd loop increase with respect to time. So, according to Lenz law a induced current develop in the IInd loop

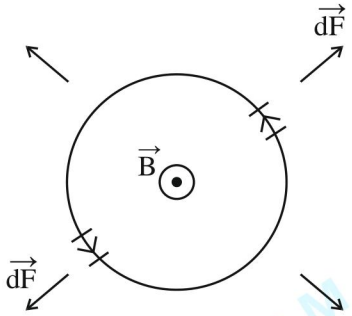


$\tau = M \times B[\text{clock wise}]$

4. For the situation shown in figure, the coil is in stretched condition.

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Therefore, the coil has a tendency to contract.



6. $\tau = pE \sin\theta$

$U = -pE \cos\theta$

τ_{max} at $\theta = 90°$

$U(\theta = 90°) = 0$

τ_{min} at $\theta° = 0°$

$U(\theta = 0°) = -pE$

7. In initial state inductor keeps its path open.

10. Conceptual

11. The current in the inductor decays gradually hence bulbs will glow even after battery is disconnected.

12. Conceptual

13. $\lambda = \dfrac{h}{mv}$

15. Conceptual

16. Conceptual

18. Speed of light is more than speed of sound

19. Conceptual

20. Conceptual

22. Fresnel distance = $Z_F < \dfrac{a^2}{\lambda} \Rightarrow \dfrac{Z_F\lambda}{a^2} < 1$

25. For $\omega > \dfrac{1}{\sqrt{LC}}$, circuit is inductive hence, current lags behind voltage.

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CHEMISTRY

26. Conc. H_2SO_4 is used in brown ring test.

27. Electron transfer takes place from anode to cathode and current flows from cathode to anode

28. Arrhenius equation is applicable mostly for 1st order reactions only

29. Conceptual

30. Even though Ce^{+4} is favoured by its noble gas configuration, it is strong oxidant, reverting to common oxidation state of +3. E^0 of Ce^{4+}/Ce^{+3}= 1.74V suggests that Ce^{4+} can oxidize even water(but reaction is slow)

31. NCERT Lab Manual (reference)

34. $\left[Co(NH_3)_2(NO_2)_2\right]^+$ and $\left[Co(NH_3)_4(NO_2)(ONO)\right]^+$ are linkage isomers. $\left[Co(NH_3)_4(NO_2)_2\right]^+$ exhibits geometrical isomerism but both the geometrical isomers are optically inactive $\left[Co(NH_3)_4(NO_2)_2\right]Cl$ and $\left[Co(NH_3)_4(NO_2)Cl\right]NO_2$ are ionisation isomers.

35. acts as both ortho, para and meta directing group in the presence of acid due to salt formation.

36. Aldehydes are readily oxidized by mild oxidizing agents such as Tollens' reagent, whereas most aliphatic ketones do not undergo oxidation under these conditions.

37. The system shows positive deviation ⇒ A–B interactions are weaker than A–A and B–B. Such systems commonly give a minimum-boiling (positive) azeotrope (boils at a lower temperature than either pure component).
Therefore statement (A) is false — a mixture can boil lower than both pure b.p.'s.

A minimum-boiling azeotrope corresponds to a maximum of total vapour pressure at fixed temperature (so it gives the lowest boiling temperature at 1 atm). Statement (B) says "lowest vapour pressure" at the azeotrope — that is the

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opposite, so (B) is false.

For fractional distillation of a system with a minimum-boiling azeotrope: the more volatile component is removed preferentially as distillate; the residue becomes richer in the less volatile component. The statement (C) assumes you start richer in CS_2 (the less volatile component under the stated ordering) than the azeotrope — then distillation will concentrate CS_2 in the flask (you can ultimately obtain the less-volatile component as residue). So (C) is correct.

Positive deviation (weaker A–B interactions) normally means mixing is endothermic ($\Delta H_mix > 0$): heat is absorbed on mixing, so in an adiabatic vessel the temperature falls, not rises. Thus (D) is false.

Property	Behavior
Intermolecular force	$F_{A-B} < F_{A-A}$ or F_{B-B}
ΔH_{mixing}	Positive (Endothermic / Cooling effect)
ΔV_{mixing}	Positive (Expansion)
Azeotrope Type	Minimum Boiling Azeotrope
Vapor Pressure	Higher than Raoult`s law prediction

38. Conceptual

40. Asseriton(A) is true. In $\left[Fe(CN)_6\right]^{3-}$, iron is in the +3 state (Fe^{3+}, d^5).

Because CN^- is a strong field ligand, the complex is low-spin (electronic configuration $t_{2g}^5 e_g^0$) giving one unpaired electron. Spin-only $\mu = \sqrt{n(n+2)}$ with $n = 1 + \mu \approx 1.73BM$.

Reason(R) is false. CN^- is not a π-donor - it is a π-acceptor (π-backbonding) ligand. As a π-acceptor, CN^- stabilizes the t_{2g} orbitals and increases Δ_0, promoting pairing. R incorrectly calls CN^- a π-donor and claims it reduces Δ_0; both parts are wrong.

Therefore A true and R false.

41. In aqueous solution, Cr^{3+} (d^3) has exceptionally high stability because in an

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octahedral field the t_{2g} set is exactly half-filled, giving large crystal field stabilization energy (CFSE).

$\Rightarrow$ Hence Cr^{2+} readily oxidizes to Cr^{3+}, so Cr^{2+} acts as a strong reducing agent.

Mn^{3+} (d^4) is comparatively unstable in water, while Mn^{2+} (d^5) has a half-filled, highly stable configuration.

$\Rightarrow$ Mn^{3+} readily gets reduced to Mn^{2+}, so Mn^{3+} acts as a strong oxidizing agent.

43. Only hydrogen atoms bonded to oxygen are ionizable; H_3PO_3 has only two – OH groups, so it is dibasic, not tribasic.

44. VOX_3 exist

X = F, Cl, Br, I

45. or an ideal solution (Raoult's law):

$$y_B = \frac{(x_B P_B^0)}{(x_A P_A^0 + x_B P_B^0)}$$

Since $P_B^0 > P_A^0$,

B is more volatile

Vapour phase is always richer in B than liquid phase

Hence, $y_B > x_B$ for all $0 < x_B < 1$

46. Conceptual

47. (i) Brown ring test $\Rightarrow$ N present.

(ii) Black ppt with lead(II) acetate $\Rightarrow$ PbS $\Rightarrow$ S present; cation Pb^{2+} belongs to Group II.

(iii) $AgNO_3$ gives white ppt soluble in NH_3 $\Rightarrow$ AgCl $\Rightarrow$ Cl present.

So, extra elements confirmed: N, S, Cl $\Rightarrow$ X = 3.

Group of cation forming black ppt: Pb^{2+} $\Rightarrow$ Y = 2.

48. In all seven cases, either precipitation or complex formation reduces the total number of solute particles in solution. Since

$$\Delta T_b \propto \text{total particle concentration } (i \times m)$$

the final solution always has fewer effective particles than the initial state, so the boiling point elevation decreases in every case.

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Hence, the number of cases where $\Delta T_{b.final} > \Delta T_{b.initial}$ is zero.

49. $\mu = \sqrt{15} \Rightarrow 3$ unpaired electrons in M^{x+}

$(3d + 4s) = 6 \Rightarrow M = Cr\left(3d^5 4s^1\right)$

3 unpaired in cation $\Rightarrow x=3\left(Cr^{3+} : 3d^3\right)$

Neutral Cr has 6 unpaired electrons $\Rightarrow y=6$

$\therefore\ x+y=9$

MATHEMATICS

51. $I.F = e^{\int \frac{xdx}{x^2-1}} = \sqrt{1-x^2}$

$$y = f(x) = \frac{3x^2 + x^4 + 2x^2 + x}{\sqrt{1-x^2}} = \int_{-1/2}^{1/2} \frac{f(x)}{\left(x^2+2\right)} dx = \frac{\pi}{6} - \frac{1}{4}$$

52. $f(x) = \dfrac{1}{x^2 + 2(a-3)x + (a-1)}$ is continuous for disc. < 0

$\Rightarrow 4(a-3)^2 - 4(a-1) < 0$

$\Rightarrow a^2 - 6a + 9 - a + 1 < 0$

$\Rightarrow a^2 - 7a + 10 < 0 \Rightarrow (a-2)(a-5) < 0$

$\Rightarrow a \in (2,5)$

$\therefore$ for $2 < a < 5$, $\dfrac{1}{x^2 + 2(a-3)x + (a-1)}$ is continuous and hence integrable

53. $x = P(A \text{ wins}) = P(H) + P(TH) + P(TTTH) + P(TTTTH) + \ldots$

$= p + qp + q^3p + q^4p + \ldots. (q = 1-p)$

$= \left(p + q^3p + q^6p + \ldots\right) + \left(qp + q^4p + q^7p + \ldots.\right)$

$= \dfrac{p}{1-q^3} + \dfrac{pq}{1-q^3} = \dfrac{1+q}{1-q+q^2}$ and $y = 1 - x = \dfrac{q^2}{1-q+q^2}$

Now, $0 < p < 1 \Rightarrow 0 < q < 1$, so, $1 + q > q^2 \Rightarrow x > y$

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Also, $x > y \Rightarrow 2x > x + y = 1$ or $x > \frac{1}{2}$ thus $y < \frac{1}{2}$

54. Definition of ellipse

55. $$\int \frac{x^2\left(x^6 + x^5 - 1\right)}{\left(2x^6 + 3x^5 + 2\right)^2} dx$$

$$= \int \frac{x^2\left(x^6 + x^5 - 1\right)}{x^6\left(2x^3 + 3x^2 + \frac{2}{x^3}\right)^2} dx$$

$$= \int \frac{\left(x^2 + x - x^{-4}\right)dx}{\left(2x^3 + 3x^2 + \frac{2}{x^3}\right)^2}$$

57. Statement 2 is correct, check statement 1 using 2

58. II, III are correct

$$N = Q - P = -1^2 + 25^2 + 2^2\left(1^2 - 3^2\right) + 4^2\left(3^2 - 5^2\right) + ... + 24^2\left(23^2 - 25^2\right)$$

$$N = 624 - 2^5\left(1^3 + 2^3 + ... + 12^3\right) < 0$$

$$|N| + 624 = 2^5\left(6 \times 13\right)^2 = 2^7.3^2.13^2$$

59. III, IV are correct

60. Put $x = i$ in $(1+x)^n = {}^nC_0 + {}^nC_1 + ... + {}^nC_n x^n$

and observe given series is real part of LHS

61. Bayes theorem

62. I, III, IV are correct

63. Conceptual

64. $$2| = 2\int_{\pi/8}^{3\pi/8} \frac{11 + \cos 4x}{1 - \cos 4x} dx$$

$$\Rightarrow |=| = \int_{\pi/8}^{3\pi/8} \frac{12 - (1 - \cos 4x)}{1 - \cos 4x} = 12\int_{\pi/8}^{3\pi/8} \frac{1}{\sin^2 2x} dx - \int_{\pi/8}^{3\pi/8} dx = 6\int_{\pi/8}^{3\pi/8} \operatorname{cosec}^2 2x \, dx - \frac{\pi}{4}$$

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65. To find the nummber of bijective functins $f : A \to A$ such that $f(1)+f(2)+f(3)=3$, the only possibility for this sum is 0 + 1 + 2 = 3.

$\Rightarrow$ Elements 1,2,3 in the domian can be mapped with 0,1,2 only. Therefore, number of bijective functions = 3! x 5! = 6 x 120 = 720.

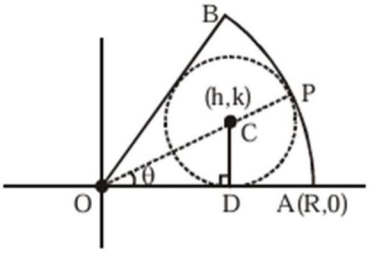
66. Let no. be x and y

$$\frac{x+y}{2}=a \text{ and } x\cdot\left(\frac{y}{x}\right)^{\frac{1}{3}}=b,\ x\left(\frac{y}{x}\right)^{\frac{2}{3}}=c$$

67. $P(H_i \cap A_m)=\frac{{}^{9}C_{m-1}}{2^{10}}$ for H_i and A_m

68. I, III are correct

Let the radius of the inscribed circle be 'r'



$\Rightarrow OC = R-r$ $CD=(R-r)\sin\theta = r-k$

Also $h=(R-r)\cos\theta; k=(R-r)\sin\theta$

$\Rightarrow h^2+k^2=(R-k)^2 \Rightarrow$ locus is $x^2=-2R\left(y-\frac{R}{2}\right)$

69. II, IV are correct

$x^{13}+2x^{12}+3x^{11}+....+13x+14=f(x)$

$$f(x)=\frac{x^{15}-15x+14}{(x-1)^2},\ \alpha=\text{cis}\left(\frac{2\pi}{15}\right)$$

$f(\alpha)f(\alpha^2).....f(\alpha^{14})=15^{13}=3^3.5^{13}$

Number of divisors = (13 + 1)(13 + 1) = 196

Number of divisors that are perfect squares = (6 + 1)(6 + 1) = 49

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Number of divisors that are perfect cubes = (4 + 1)(4 + 1) = 25

70. Conceptual

73. Count manually by assigning the values to z, then y then x.

74. $$\int_0^2 (1+x)^n \, dx = \int_0^2 \left({}^nC_0 + {}^nC_1 x + \ldots\ldots + {}^nC_n x^n \right) dx$$

$$\frac{3^{n+1}}{n+1} - \frac{1}{n+1} = {}^nC_0 \cdot 2 + {}^nC_1 \cdot \frac{2^2}{2} + \ldots\ldots\ldots$$

$n = 20$

$$\frac{3^{21}}{21} - \frac{1}{21} = 2\left({}^{20}C_0 + {}^{20}C_1 + \ldots\ldots \right)$$

$${}^{20}C_0 + {}^{20}C_1 + \ldots\ldots = \frac{3^{21} - 1}{42}$$

75. Conceptual

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